

Coordinate Geometry

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Plotting Points

Along the hall, then up the stairs. Type coordinates like 3,5.

Warm-Up 12 ITEMS • RECALL

1 In (3, 5), the number along

2 In (3, 5), the number up

3 The origin — type it as a,b

4 4 along, 2 up — type as a,b

5 0 along, 6 up — type as a,b

6 In (7, 1), the x value

7 In (4, 7), the number across

8 In (4, 7), the number up

9 The origin — type as a,b

10 6 across, 3 up — type as a,b

11 0 across, 5 up — type as a,b

12 In (9, 2), the first number

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Distance from (2,3) to (7,3)

.....

14. Distance from (4,1) to (4,8)

.....

15. (1,2), (1,6), (5,6), (5,2) — how many sides

.....

16. That shape's side length along the bottom

.....

17. A point 3 right of (2,5) — type as a,b

.....

18. 4 right of (3,6) — type as a,b

.....

19. 3 up from (5,2) — type as a,b

.....

20. Distance from (1,4) to (8,4)

.....

21. Distance from (3,2) to (3,9)

.....

22. Halfway between (0,0) and (10,0) — type as a,b

.....

★ Challenge NOT GRADED • REASON IT OUT

23. (1,1), (1,5), (6,5), (6,1) — the area

.....

24. Perimeter of that rectangle

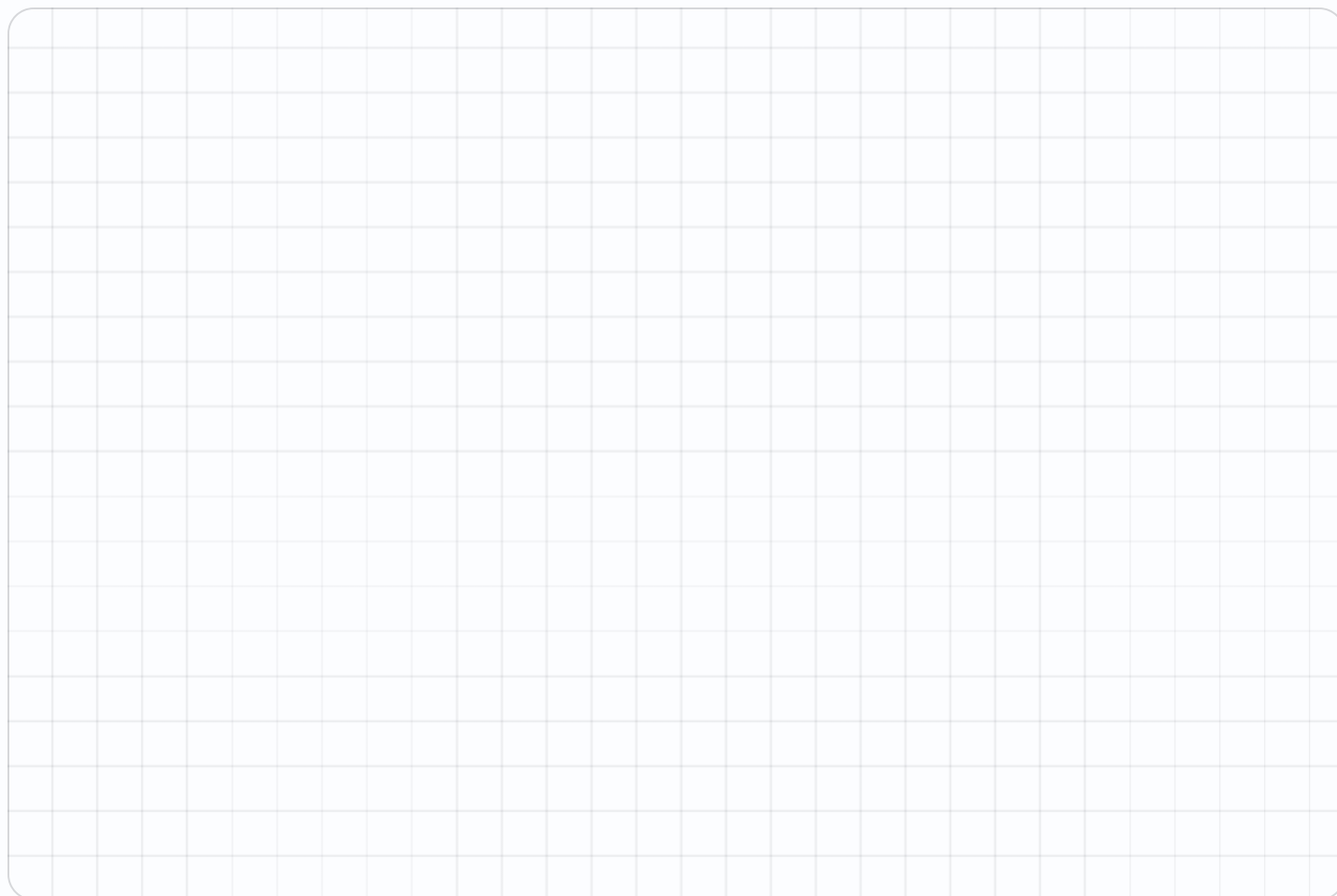
.....

25. Halfway between (2,4) and (10,8) — type as a,b

.....

26. From (0,0) to (7,9) across and up — total steps

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Patterns as Points

Two rules, paired term by term.

Warm-Up 12 ITEMS • RECALL

1 Rule + 3 from 0: 0, 3, 6, ?

2 Rule + 6 from 0: 0, 6, 12, ?

3 Rule + 2 from 1: 1, 3, 5, ?

4 Rule + 5 from 0 — the third term

5 Rule + 4 from 2 — the fourth term

6 Rule + 10 from 0 — the fifth term

7 Rule $\times 2$: in 3 — the pair as a,b

8 Rule $\times 2$: in 5

9 Rule + 3: in 4

10 Rule $\times 3$: in 2

11 Rule + 1: in 9

12 Rule $\times 4$: in 3

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Rules +3 and +6 from 0. Second is how many times the first
.....

14. Pair the third terms of +3 and +6 from 0 — type as a,b
.....

15. Rules +2 and +4 from 0 — pair the fourth terms as a,b
.....

16. Rule + 5 from 0 — the tenth term
.....

17. Rule $\times 2$ from 1 — the fifth term
.....

18. Rule $\times 2 + 1$: in 5 — the pair as a,b
.....

19. Rule $\times 3 - 2$: in 4
.....

20. Rule $\times 5$: in 6
.....

21. Rule $\times 2$: which input gives 14
.....

22. Rule $\times 3 + 1$: which input gives 16
.....

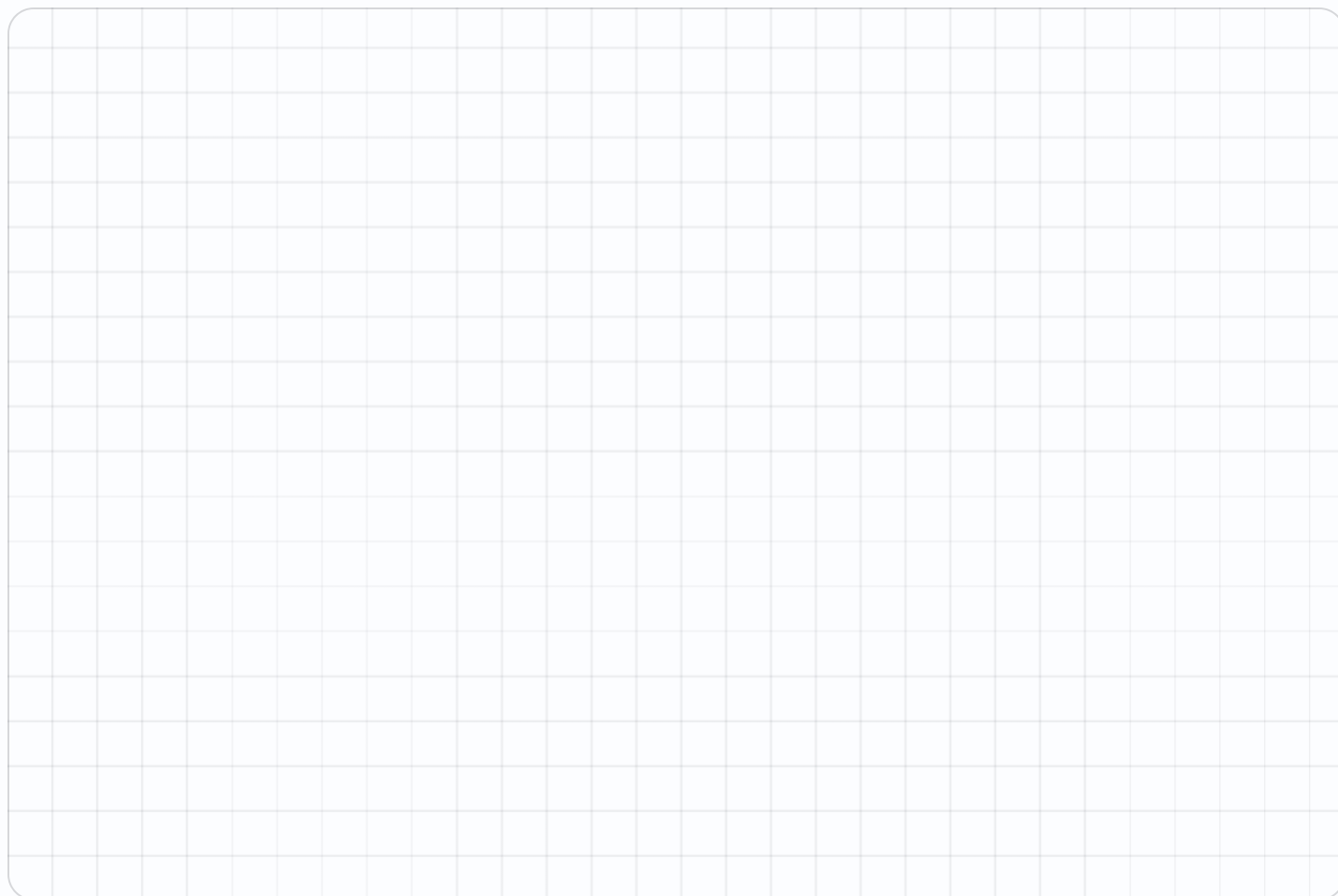
★ Challenge NOT GRADED • REASON IT OUT

23. Rules +3 and +9 — second is how many times the first

24. Rule + 7 from 3 — the twentieth term

25. Rule $\times 4 - 3$: in 7 — the pair as a,b

26. A steady rule always plots as a — type line or curve

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Quadrilaterals

Properties first, names second.

Warm-Up 9 ITEMS • RECALL

1 Sides on a quadrilateral

2 Right angles in a rectangle

3 Equal sides on a square

4 Pairs of parallel sides in a parallelogram

5 Sides on a triangle

6 Right angles in a square

7 Angles in a quadrilateral add to

8 Pairs of parallel sides in a trapezoid

9 Sides on a rhombus

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. Is every square a rectangle — yes or no

.....

11. Is every rectangle a square — yes or no

.....

12. Is every rectangle a parallelogram — yes or no

.....

13. Angles in any triangle add to

.....

14. Angles in any quadrilateral add to

.....

15. Is every square a rhombus — yes or no

.....

16. Is every rhombus a square

.....

17. Is every rectangle a parallelogram

.....

18. A rhombus with right angles is a

.....

19. Equal sides on a rhombus

.....

Challenge

20. A quadrilateral with exactly one pair of parallel sides — type its name

21. A rhombus with right angles is also called a

22. A quadrilateral with exactly one pair of parallel sides

23. The most specific name for a 4-sided shape with 4 equal sides and 4 right angles



Working space



Error journal

Always, Sometimes, Never

Judge the statement, then justify it out loud.

Warm-Up 12 ITEMS • RECALL

1 A square is a rectangle —
always, sometimes or never

2 A rectangle is a square

3 A triangle has four sides

4 A parallelogram has
parallel sides

5 A quadrilateral has a right
angle

6 A square is a
quadrilateral

7 Is every square a
rectangle — yes or no

8 Is every rectangle a
square

9 Is every square a
quadrilateral

10 Do all triangles have 3
sides

11 Do all rectangles have 4
right angles

12 Is a circle a polygon

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A rhombus is a square

14. A trapezoid is a parallelogram

15. A rectangle's diagonals are equal

16. A triangle has two right angles

17. A parallelogram is a rectangle

18. A rectangle is sometimes a square — type always, sometimes or never

19. A triangle is a quadrilateral

20. A rhombus is a parallelogram

★ Challenge NOT GRADED • REASON IT OUT

21. Every rhombus is a parallelogram

.....

22. A shape with four equal sides is a square

.....

23. A parallelogram is a rhombus

.....

24. An equilateral triangle is isosceles

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Map My Block

Enrichment. Real places, ordered pairs.

✦ Warm-Up 4 ITEMS • RECALL

1 Landmarks on a ten-stop map

2 The origin of your map — type as a,b

3 Distance from (2,2) to (7,2)

4 Distance from (4,1) to (4,6)

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. From (2,3), move 4 right and 1 up — type as a,b
.....

6. Blocks from (1,1) to (1,9)
.....

7. A route (1,1) to (6,1) to (6,5) — total
.....

8. (2,2),(2,6),(7,6),(7,2) — the area
.....

★ Challenge NOT GRADED • REASON IT OUT

9. From (0,0) to (6,8) going only along and up — total blocks

.....

10. A park at (3,4) and a school at (9,4) — blocks apart

.....

11. Rule: each block is 2 units. (5,5) is how many units from the origin along x

.....

12. That shape's perimeter

.....

13. Halfway between (1,1) and (9,7) — type as a,b

.....

14. A square with (3,3) and (8,8) opposite — its area

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Plotting Points

OUT OF 22

Warm-Up • 1-12 • 3 | 5 | 0,0 | 4,2 | 0,6 | 7 | 4 | 7 | 0,0 | 6,3 | 0,5 | 9

Core • 13-22 • 5 | 7 | 4 | 4 | 5,5 | 7,6 | 5,5 | 7 | 7 | 5,0

Challenge • 23-26 • 20 | 18 | 6,6 | 16

1.2 • Patterns as Points

OUT OF 22

Warm-Up • 1-12 • 9 | 18 | 7 | 10 | 14 | 40 | 3,6 | 5,10 | 4,7 | 2,6 | 9,10 | 3,12

Core • 13-22 • 2 | 6,12 | 6,12 | 45 | 16 | 5,11 | 4,10 | 6,30 | 7 | 5

Challenge • 23-26 • 3 | 136 | 7,25 | line

1.3 • Quadrilaterals

OUT OF 19

Warm-Up • 1-9 • 4 | 4 | 4 | 2 | 3 | 4 | 360 | 1 | 4

Core • 10-19 • yes | no | yes | 180 | 360 | yes | no | yes | square | 4

Challenge • 20-23 • trapezoid | square | trapezoid | square

1.4 • Always, Sometimes, Never

OUT OF 20

Warm-Up • 1-12 • always | sometimes | never | always | sometimes | always | yes | no | yes | yes | yes | no

Core • 13-20 • sometimes | sometimes | always | never | sometimes | sometimes | never | always

Challenge • 21-24 • always | sometimes | sometimes | always

Fri • Map My Block

OUT OF 8

Warm-Up • 1-4 • 10 | 0,0 | 5 | 5

Core • 5-8 • 6,4 | 8 | 9 | 20

Challenge • 9-14 • 14 | 6 | 10 | 18 | 5,4 | 25

Two Rules at Once

Add 3 and add 6, side by side, from the same start.

Warm-Up 12 ITEMS • RECALL

1 Rule + 3 from 0: 0, 3, 6, ?

2 Rule + 6 from 0: 0, 6, 12, ?

3 Rule + 2 from 0 — the third term

4 Rule + 4 from 0 — the third term

5 Rule + 5 from 0 — the fourth term

6 Rule + 10 from 0 — the fourth term

7 Rule $\times 2$: in 4

8 Rule + 3: in 4

9 Rule $\times 2$: in 6

10 Rule + 3: in 6

11 Rule $\times 2$: in 10

12 Rule + 3: in 10

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Rules +3 and +6 — the second is how many times the first

.....

14. Rules +2 and +6 — how many times

.....

15. Rules +5 and +10 — how many times

.....

16. Rule + 3 from 0 — the tenth term

.....

17. Rule + 6 from 0 — the tenth term

.....

18. Rules $\times 2$ and $\times 2 + 3$ — the gap at any input

.....

19. Rules + 5 and + 9 — the gap

.....

20. Rule $\times 3$ vs $\times 3 + 2$ — the gap

.....

21. Rule $\times 2$ at 8 minus rule + 3 at 8

.....

22. Rule $\times 4$ at 5 minus rule $\times 2$ at 5

.....



Working space



Error journal

Make the Ordered Pairs

Pair the terms in order. First rule across, second up.

Warm-Up 12 ITEMS • RECALL

1 First terms of +3 and +6 from 0 — type as a,b

2 Second terms — type as a,b

3 Third terms — type as a,b

4 Fourth terms — type as a,b

5 In (3,6), the across value

6 In (3,6), the up value

7 Rule $\times 2$: in 1 — the pair as a,b

8 In 2

9 In 3

10 Rule $\times 5$: in 2

11 Rule + 4: in 3

12 Rule $\times 3$: in 4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Rules +2 and +4, third terms — type as a,b

.....

14. Rules +5 and +10, second terms — type as a,b

.....

15. Rules +1 and +3, fourth terms — type as a,b

.....

16. In every pair from +3 and +6, up is how many times across

.....

17. Rules +2 and +6, fourth terms — type as a,b

.....

18. Rule $\times 2 + 1$: in 4 — the pair as a,b

.....

19. Rule $\times 3 - 1$: in 5

.....

20. Rule $\times 10$: in 7

.....

21. Rule $\times 2 + 5$: in 6

.....

22. Rule $\times 6$: in 8

.....

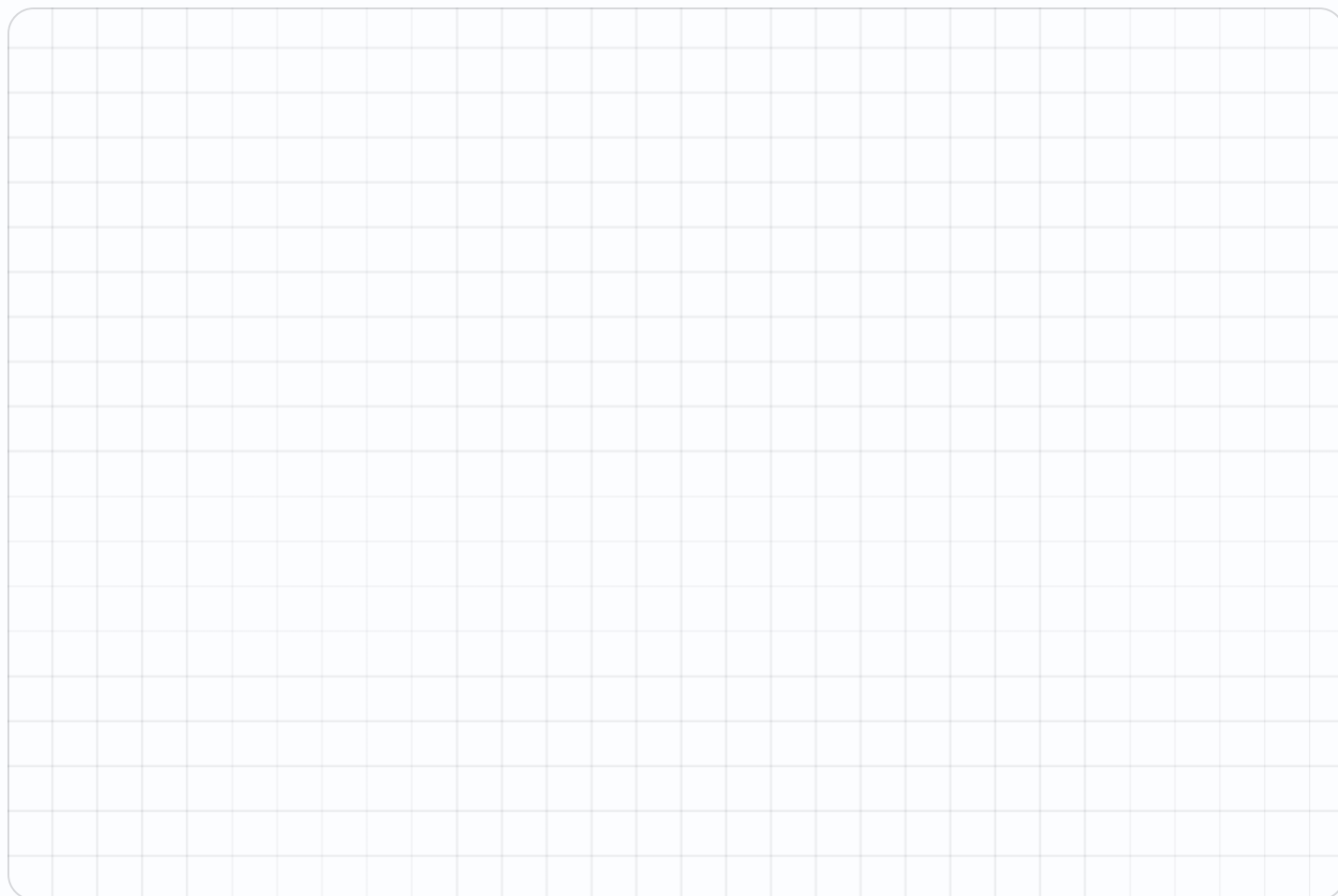
★ Challenge NOT GRADED • REASON IT OUT

23. Rules +4 and +12, third terms — type as a,b

24. If across is 15 under rules +3 and +6, the up value

25. Rule $\times 4 + 2$: in 9 — the pair as a,b

26. Rule $\times 2 - 3$: in 12

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Plot the Pairs

They land on a straight line through the origin. Say why.

Warm-Up 12 ITEMS • RECALL

1 Plot (0,0) — the across value

2 Plot (3,6) — the up value

3 Do (0,0), (3,6), (6,12) lie on a line — yes or no

4 Distance across from (0,0) to (3,6)

5 Distance up from (0,0) to (3,6)

6 The up value when across is 6

7 Plot (2,4) then (3,6) — the up values differ by

8 Plot (1,2) then (2,4) — the across values differ by

9 A rule $\times 2$ climbs how much per step across

10 A rule $\times 3$ climbs how much per step

11 A rule $+ 5$ climbs how much per step

12 A rule $\times 5$ climbs how much per step

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. On that line, the up value when across is 9
.....

14. When across is 12
.....

15. When across is 20
.....

16. The rule linking them — type the multiplier
.....

17. When up is 30, the across value
.....

18. From (1,2) with rule $\times 2$ — the pair at 4 as a,b
.....

19. A line through (0,0) and (3,6) — the up value at 5
.....

20. A line through (0,3) climbing 2 — the up value at 4
.....

21. A rule $\times 2 + 3$ at input 0 — the up value
.....

22. A steeper line means a bigger — type multiplier or adder
.....

★ Challenge NOT GRADED • REASON IT OUT

23. Rules +2 and +6 — the multiplier linking them

24. On that line, up when across is 14

25. A line through (0,1) and (4,9) — its climb per step

26. That line's up value at 10

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Relate the Sequences

Every second term is double the first. Prove it three times.

Warm-Up 12 ITEMS • RECALL

1 Double 3

2 Double 6

3 Double 9

4 Half of 12

5 Half of 18

6 Triple 4

7 Sequence $\times 2$ from 1: the third term

8 Sequence $+ 3$ from 1: the third term

9 Sequence $\times 2$ from 1: the fifth term

10 Sequence $+ 3$ from 1: the fifth term

11 2, 4, 6, 8 — the next

12 1, 4, 7, 10 — the next

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Rules $+3$ and $+6$, the fifth terms — type the second one

.....

14. Rules $+4$ and $+8$, the third terms — type the second one

.....

15. If the first sequence reaches 21, the second reaches

.....

16. Rules $+5$ and $+15$ — the multiplier

.....

17. Rules $+2$ and $+8$ — the multiplier

.....

18. Sequences 2,4,6,8 and 5,7,9,11 — the constant gap

.....

19. Sequences 3,6,9 and 6,12,18 — the second is how many times the first

.....

20. Sequence 0,2,4,6 and 0,4,8,12 — the ratio

.....

21. 2,4,6,8 paired with 4,8,12,16 — the pair at term 3 as a,b

.....

22. Two sequences differing by 5 stay apart by

.....

Rule Race

One plots, the other names the rule.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up 4 ITEMS • RECALL

1 Pairs (1,2), (2,4), (3,6) — the multiplier

2 Pairs (1,3), (2,6) — the multiplier

3 Rule $\times 2$: in 7

4 Rule $+ 6$: in 7

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Pairs (1,5), (2,10), (3,15) — up when across is 6

6. Pairs (2,8), (3,12) — the multiplier

7. Rule $\times 3 + 2$: in 8

8. Rule $\times 4 - 1$: in 6

 Challenge NOT GRADED · REASON IT OUT

9. Pairs (1,4), (2,7), (3,10) — the rule is $\times 3 + ?$ — type the add

10. That rule — up when across is 10

11. Pairs (2,9), (3,13) — the multiplier

12. Rule $\times 5 + 3$: in 12

13. Rule $\times 2$ and $\times 2 + 9$ — the gap

14. A rule giving $5 \rightarrow 17$ and $8 \rightarrow 26$ — its multiplier

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Two Rules at Once

OUT OF 22

Warm-Up • 1-12 • 9 | 18 | 4 | 8 | 15 | 30 | 8 | 7 | 12 | 9 | 20 | 13

Core • 13-22 • 2 | 3 | 2 | 27 | 54 | 3 | 4 | 2 | 5 | 10

Challenge • 23-26 • 3 | 136 | parallel | 10

2.2 • Make the Ordered Pairs

OUT OF 22

Warm-Up • 1-12 • 0,0 | 3,6 | 6,12 | 9,18 | 3 | 6 | 1,2 | 2,4 | 3,6 | 2,10 | 3,7 | 4,12

Core • 13-22 • 4,8 | 5,10 | 3,9 | 2 | 6,18 | 4,9 | 5,14 | 7,70 | 6,17 | 8,48

Challenge • 23-26 • 8,24 | 30 | 9,38 | 12,21

2.3 • Plot the Pairs

OUT OF 22

Warm-Up • 1-12 • 0 | 6 | yes | 3 | 6 | 12 | 2 | 1 | 2 | 3 | 1 | 5

Core • 13-22 • 18 | 24 | 40 | 2 | 15 | 4,8 | 10 | 11 | 3 | multiplier

Challenge • 23-26 • 3 | 42 | 2 | 21

2.4 • Relate the Sequences

OUT OF 22

Warm-Up • 1-12 • 6 | 12 | 18 | 6 | 9 | 12 | 4 | 7 | 16 | 13 | 10 | 13

Core • 13-22 • 24 | 16 | 42 | 3 | 4 | 3 | 2 | 2 | 6,12 | 5

Challenge • 23-26 • $\frac{3}{2}$ | 36 | 3 | 3,9

Fri • Rule Race

OUT OF 8

Warm-Up • 1-4 • 2 | 3 | 14 | 13

Core • 5-8 • 30 | 4 | 26 | 23

Challenge • 9-14 • 1 | 31 | 4 | 63 | 9 | 3

Properties of Quadrilaterals

Sides, angles, parallels. Name it from what it does.

Warm-Up 9 ITEMS • RECALL

1 Sides on a quadrilateral

2 Right angles in a rectangle

3 Equal sides on a square

4 Pairs of parallel sides in a parallelogram

5 Right angles in a square

6 Equal sides on a rhombus

7 Sides on a parallelogram

8 Angles in a quadrilateral add to

9 Pairs of parallel sides in a rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. Pairs of parallel sides in a trapezoid

.....

11. Angles in any quadrilateral add to

.....

12. A quadrilateral with three 90° angles — the fourth

.....

13. Equal opposite sides on a parallelogram — how many pairs

.....

14. Lines of symmetry in a rectangle

.....

15. Opposite angles in a parallelogram are — type equal or different

.....

16. A parallelogram with a 70° angle — its neighbour

.....

17. Diagonals of a rectangle are — type equal or different

.....

18. A rhombus's diagonals meet at how many degrees

.....

★ Challenge NOT GRADED • REASON IT OUT

19. A quadrilateral with exactly one pair of parallel sides

.....

20. A rhombus with right angles is also a

.....

21. A parallelogram with angles 60, 120, 60 — the fourth

.....

22. A kite has how many pairs of equal adjacent sides

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

The Hierarchy

Every square is a rectangle. Not the other way round.

Warm-Up 10 ITEMS • RECALL

1 Is every square a rectangle — yes or no

2 Is every rectangle a square

3 Is every square a quadrilateral

4 Is every quadrilateral a square

5 Is a square a rhombus

6 Is every rhombus a square

7 Is a square a rectangle — yes or no

8 Is a rectangle a square

9 Is a rhombus a parallelogram

10 Is a parallelogram a quadrilateral

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Is every rectangle a parallelogram — yes or no

.....

12. Is every parallelogram a rectangle

.....

13. Is every rhombus a parallelogram

.....

14. Is a trapezoid a parallelogram

.....

15. Is every square a parallelogram

.....

16. The most specific name for a rhombus with right angles

.....

17. The most specific name for a parallelogram with right angles and unequal sides

.....

18. Rings that contain the square, counting quadrilateral, parallelogram, rectangle and rhombus

.....

19. Is every rectangle a parallelogram

.....

Classify Triangles

By sides and by angles. Two answers for every triangle.

Warm-Up 7 ITEMS • RECALL

1 Sides on a triangle

2 Angles in a triangle add to

3 Equal sides on an equilateral triangle

4 Equal sides on an isosceles triangle

5 Equal sides on a scalene triangle

6 Each angle of an equilateral triangle

7 Right angles in a right triangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. A triangle with a 90° angle — type its name

.....

9. A triangle with 90° and 30° — the third angle

.....

10. A triangle with 40° and 60° — the third

.....

11. An isosceles triangle with a 40° apex — each base angle

.....

12. Can a triangle have two right angles — yes or no

.....

13. A triangle with 90° and 45° — the third angle

.....

14. A triangle with 100° and 40° — the third

.....

15. A triangle with angles 30, 60, 90 — type acute, right or obtuse

.....

16. A triangle with angles 20, 30, 130 — its type

.....

★ Challenge NOT GRADED • REASON IT OUT

17. A triangle with all angles under 90° — type its name

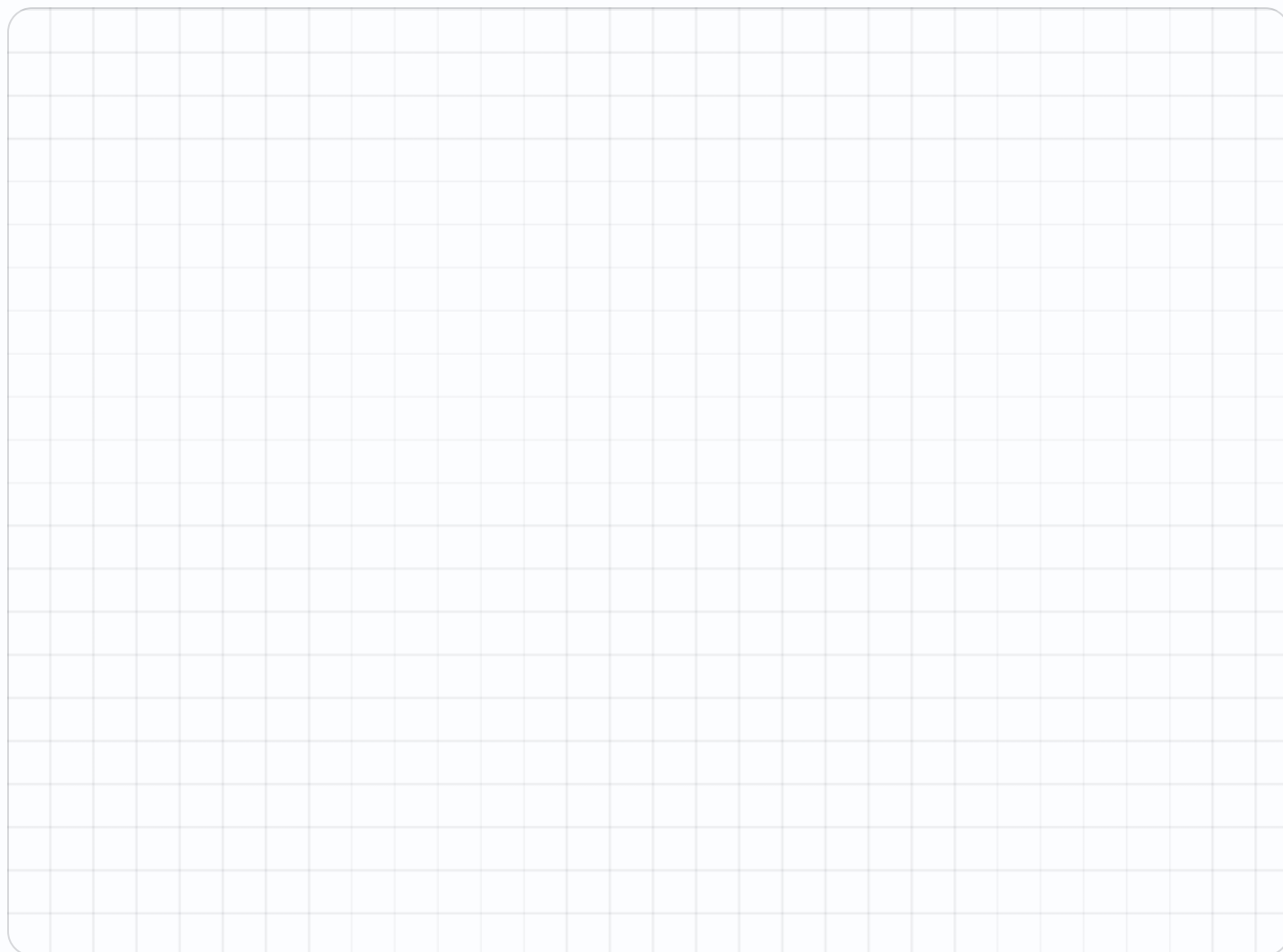
.....

18. An isosceles right triangle — each of its other two angles

.....

19. An isosceles triangle with an 80° apex — each base angle

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Always, Sometimes, Never

Judge the statement, then justify it out loud.

Warm-Up 9 ITEMS • RECALL

1 A square is a rectangle

2 A rectangle is a square

3 A triangle has four sides

4 A parallelogram has parallel sides

5 A quadrilateral has a right angle

6 A square is a quadrilateral

7 A square is a rectangle — always, sometimes or never

8 A triangle is a quadrilateral

9 A rhombus is a parallelogram

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. A rhombus is a square

.....

11. A trapezoid is a parallelogram

.....

12. A rectangle's diagonals are equal

.....

13. A triangle has two right angles

.....

14. A parallelogram is a rectangle

.....

15. An equilateral triangle is isosceles — always, sometimes or never

.....

16. An isosceles triangle is equilateral

.....

17. A right triangle is equilateral

.....

18. A trapezoid is a quadrilateral

.....

★ Challenge NOT GRADED • REASON IT OUT

19. Every rhombus is a parallelogram

.....

20. A shape with four equal sides is a square

.....

21. A scalene triangle is isosceles

.....

22. A quadrilateral with four equal sides is a square

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 In (3, 5), the across value

2 Sides on a quadrilateral

3 In (5,3), the up value

4 Angles in a quadrilateral add to

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Distance from (2,3) to (7,3)

.....

6. Rules +3 and +6, third terms — type as a,b

.....

7. Angles in any quadrilateral add to

.....

8. Is every square a rectangle — yes or no

.....

9. A triangle with 90° and 30° — the third angle

.....

10. Rule $\times 3 + 2$: in 6

.....

11. Distance from (2,5) to (9,5)

.....

12. A triangle with 90° and 30° — the third

.....

13. Is every square a rhombus — yes or no

.....

14. A rectangle is a square — always, sometimes or never

.....

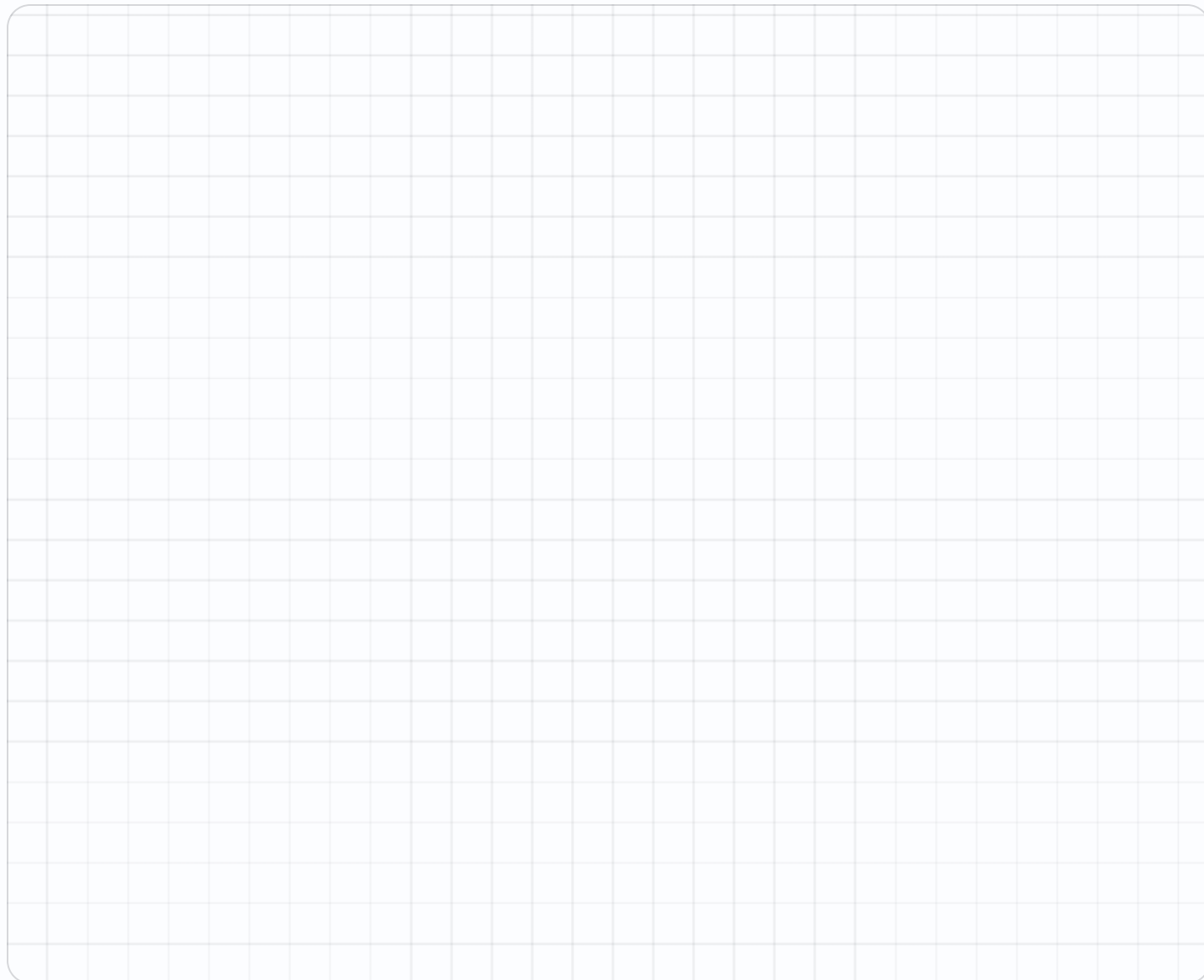
★ Challenge NOT GRADED • REASON IT OUT

15. $(1,1), (1,5), (6,5), (6,1)$ — the area

.....

16. $(1,1), (1,6), (7,6), (7,1)$ — the area

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Properties of Quadrilaterals

OUT OF 18

Warm-Up • 1-9 • 4 | 4 | 4 | 2 | 4 | 4 | 4 | 360 | 2

Core • 10-18 • 1 | 360 | 90 | 2 | 2 | equal | 110 | equal | 90

Challenge • 19-22 • trapezoid | square | 120 | 2

3.2 • The Hierarchy

OUT OF 19

Warm-Up • 1-10 • yes | no | yes | no | yes | no | yes | no | yes | yes

Core • 11-19 • yes | no | yes | no | yes | square | rectangle | 4 | yes

Challenge • 20-23 • square | quadrilateral | square | no

3.3 • Classify Triangles

OUT OF 16

Warm-Up • 1-7 • 3 | 180 | 3 | 2 | 0 | 60 | 1

Core • 8-16 • right | 60 | 80 | 70 | no | 45 | 40 | right | obtuse

Challenge • 17-19 • acute | 45 | 50

3.4 • Always, Sometimes, Never

OUT OF 18

Warm-Up • 1-9 • always | sometimes | never | always | sometimes | always | always | never
| always

Core • 10-18 • sometimes | sometimes | always | never | sometimes | always | sometimes |
never | always

Challenge • 19-22 • always | sometimes | never | sometimes

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 3 | 4 | 3 | 360

Core • 5-14 • 5 | 6,12 | 360 | yes | 60 | 20 | 7 | 60 | yes | sometimes

Challenge • 15-16 • 20 | 30

Finish the Map

Ten landmarks, ten ordered pairs, no ambiguity.

Warm-Up 11 ITEMS • RECALL

1 Landmarks on the map

2 The origin — type as a,b

3 3 across, 4 up — type as a,b

4 In (6,2), the up value

5 Distance from (0,0) to (5,0)

6 Distance from (0,0) to (0,8)

7 A landmark at (3,5) — the across value

8 Distance from (1,2) to (6,2)

9 Distance from (4,3) to (4,9)

10 5 across, 7 up — type as a,b

11 Blocks from (0,0) to (3,3) across and up

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. From (2,3), 4 right and 1 up — type as a,b

13. Blocks from (1,1) to (1,9)

14. A park at (3,4), a school at (9,4) — blocks apart

15. From (0,0) to (6,8) across-and-up — blocks

16. Halfway between (0,0) and (10,4) — type as a,b

17. A route (1,1) to (8,1) to (8,6) — total blocks

18. A square park (2,2) to (7,7) — its area

19. That park's perimeter

20. Halfway between (0,0) and (12,6) — type as a,b

21. Ten landmarks each 4 blocks apart in a line — total length

★ Challenge NOT GRADED • REASON IT OUT

22. Each block is 2 units — (5,5) is how many units across from the origin

23. A square route (1,1),(7,1),(7,7),(1,7) — total distance

24. A rectangular lake from (2,1) to (12,6) — its area

25. A patrol (1,1),(9,1),(9,5),(1,5) and back — total distance

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Write the Directions

Somebody else must be able to follow them exactly.

Warm-Up 11 ITEMS • RECALL

1 From (0,0) move 4 across — type as a,b

2 Then 3 up — type as a,b

3 Total distance

4 From (2,2) move 2 up — type as a,b

5 From (2,2) move 2 across — type as a,b

6 Are (2,4) and (4,2) the same — yes or no

7 Then 6 up — type as a,b

8 Total blocks travelled

9 From (5,5) move 3 up — type as a,b

10 From (5,5) move 3 across — type as a,b

11 Blocks from (0,0) to (2,5)

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. (1,1) to (1,7) to (5,7) — total distance

13. (2,2) to (8,2) to (8,9) — total distance

14. A round trip (0,0) to (6,0) and back

15. (3,3) to (3,10) — distance

16. A square patrol of side 5 — total distance

17. (1,2) to (1,8) to (6,8) — total blocks

18. (3,3) to (10,3) to (10,8) — total

19. A round trip (0,0) to (7,0) and back

20. (2,4) to (2,12) — the distance

21. A square patrol of side 6 — total distance

★ Challenge NOT GRADED • REASON IT OUT

22. A patrol (1,1),(7,1),(7,5),(1,5) and back — total distance

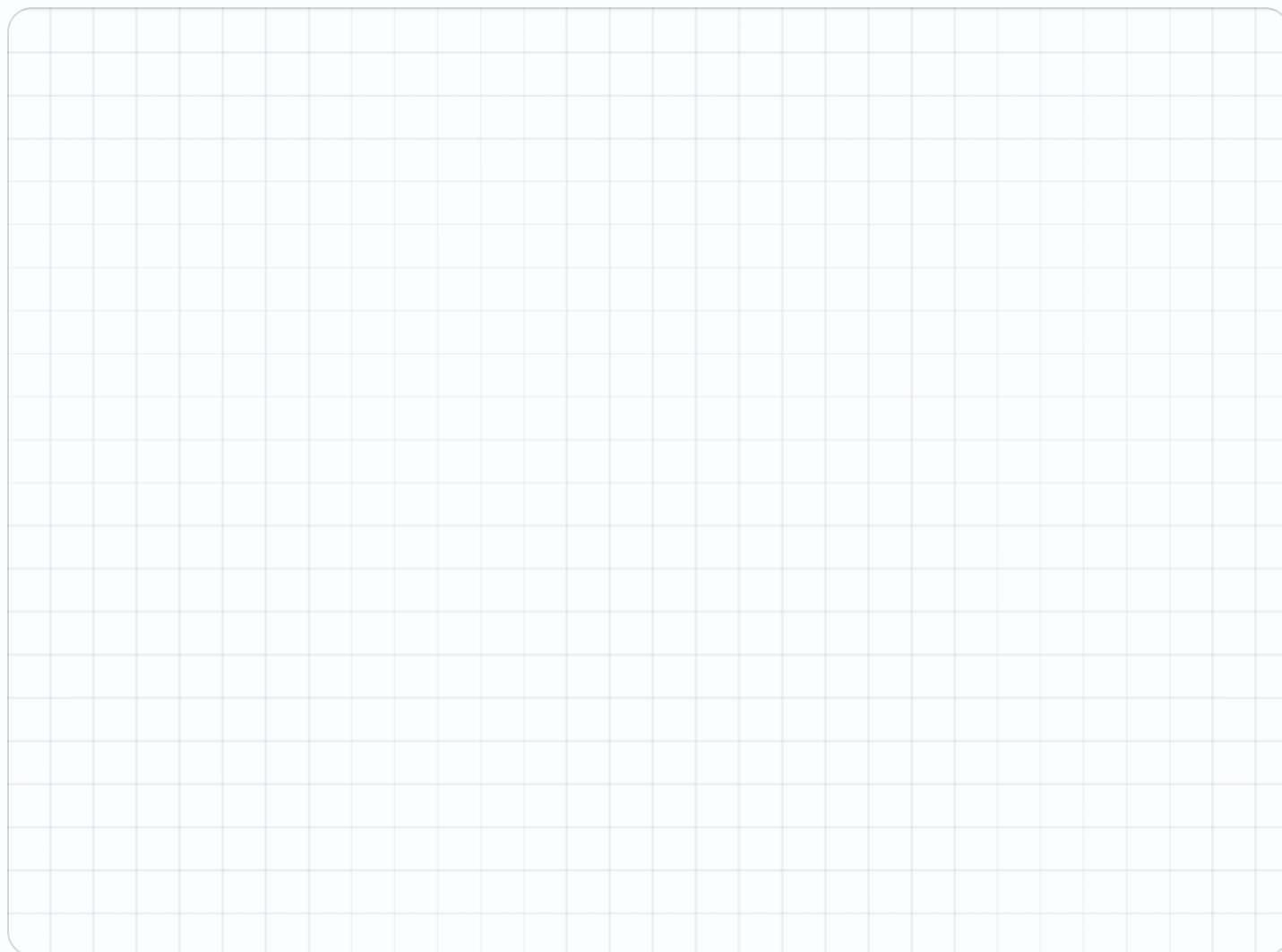
.....

23. The area enclosed by that patrol

.....

24. A patrol (2,2),(8,2),(8,7),(2,7) and back

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Walk Somebody Through It

Fix any pair that misleads them. Mark it honestly.

Warm-Up 11 ITEMS • RECALL

1 10 stops, 8 followed correctly — the percent _____	2 10 stops, 9 correct — the percent _____	3 10 minus 7 _____	4 Half of 10 stops _____
5 10 stops, 5 correct — the percent _____	6 10 stops, 10 correct — the percent _____	7 10 stops, 8 correct — the percent _____	8 10 stops, 9 correct _____
9 10 stops, 7 correct _____	10 10 minus 6 _____	11 10 stops, 10 correct _____	

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. 4 minutes a stop, 10 stops — minutes
13. Two pairs rewritten out of 10 — the percent
14. 85% of 10 stops
15. If 9 of 10 are followable, the percent that are not
16. A 40-minute walk over 10 stops — mean minutes each
17. 12 stops at 4 minutes each — minutes
18. Two rewrites out of 12 — the percent, to the nearest whole
19. 85% of 12 stops, rounded down
20. A 48-minute walk over 12 stops — mean minutes
21. 12 stops worth 10 points each — the total

★

Challenge

NOT GRADED • REASON IT OUT

22. 12 points a stop, 10 stops, scored 96 — the percent

.....

23. To score 85% of 120 points, she needs

.....

24. Scoring 102 of 120 — the percent

.....

25. To score 90% of 120 points

.....

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Year-End Sweep

All eight error journals. Name the habit that fixed itself.

Warm-Up 10 ITEMS • RECALL

1 4.2×10

2 23×14

3 $84 \div 12$

4 $0.3 + 0.4$

5 $1/4 + 1/4$

6 A 2 by 3 by 4 box

7 $48 \div 6$

8 $0.5 + 0.2$

9 $1/2 + 1/4$ — type as a/b

10 A 2 by 3 by 4 box — volume

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

11. 0.035×100

12. 237×45

13. $4536 \div 21$

14. $1/2 + 1/3$

15. $3/4 \times 2/3$

16. 2450×36

17. $4,536 \div 21$

18. $9.2 - 4.55$

19. $2/3 \times 3/4$ — type as a/b

20. Volume 144, length 6 width 4 — the height



Mission 08 Test

Twelve items plus the completed map. Final trophy band awarded.

Warm-Up 4 ITEMS • RECALL

1 In (3, 5), the up value
_____2 Sides on a quadrilateral
_____3 In (6,2), the across value
_____4 Rule $\times 2$: in 9

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. Distance from (4,1) to (4,8)
.....6. (1,1), (1,5), (6,5), (6,1) — the area
.....7. Rules +3 and +6, fourth terms — type as a,b
.....8. Rules +2 and +8 — the multiplier
.....9. Angles in a triangle add to
.....10. A triangle with 40° and 60° — the third
.....11. Is every rectangle a parallelogram — yes or no
.....12. A rhombus is a square — always, sometimes or never
.....13. Rule $\times 3 + 4$: in 7
.....14. Distance from (3,1) to (3,10)
.....15. (2,2),(2,8),(6,8),(6,2) — the area
.....16. That shape's perimeter
.....17. A triangle with 90° and 25° — the third
.....18. Halfway between (2,2) and (10,10) — type as a,b
.....19. Rules $\times 3$ and $\times 3 + 8$ — the gap
.....

 Challenge NOT GRADED · REASON IT OUT

20. A patrol $(1,1), (7,1), (7,5), (1,5)$ — the area enclosed

21. On the line from rules +3 and +6, up when across is 14

22. A patrol $(1,1), (9,1), (9,6), (1,6)$ — the area enclosed

23. A rule giving $4 \rightarrow 14$ and $7 \rightarrow 23$ — its multiplier

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Finish the Map

OUT OF 21

Warm-Up • 1-11 • 10 | 0,0 | 3,4 | 2 | 5 | 8 | 3 | 5 | 6 | 5,7 | 6

Core • 12-21 • 6,4 | 8 | 6 | 14 | 5,2 | 12 | 25 | 20 | 6,3 | 36

Challenge • 22-25 • 10 | 24 | 50 | 24

4.2 • Write the Directions

OUT OF 21

Warm-Up • 1-11 • 4,0 | 4,3 | 7 | 2,4 | 4,2 | no | 4,6 | 10 | 5,8 | 8,5 | 7

Core • 12-21 • 10 | 13 | 12 | 7 | 20 | 11 | 12 | 14 | 8 | 24

Challenge • 22-24 • 20 | 24 | 22

4.3 • Walk Somebody Through It

OUT OF 21

Warm-Up • 1-11 • 80 | 90 | 3 | 5 | 50 | 100 | 80 | 90 | 70 | 4 | 100

Core • 12-21 • 40 | 20 | 8.5 | 10 | 4 | 48 | 17 | 10 | 4 | 120

Challenge • 22-25 • 80 | 102 | 85 | 108

Thu • Year-End Sweep

OUT OF 20

Warm-Up • 1-10 • 42 | 322 | 7 | 0.7 | $\frac{1}{2}$ | 24 | 8 | 0.7 | $\frac{3}{4}$ | 24

Core • 11-20 • 3.5 | 10665 | 216 | $\frac{5}{6}$ | $\frac{1}{2}$ | 88200 | 216 | 4.65 | $\frac{1}{2}$ | 6

Challenge • 21-24 • 4.544 | 20 | $\frac{41}{24}$ | 9,38

Fri • Mission 08 Test

OUT OF 19

Warm-Up • 1-4 • 5 | 4 | 6 | 18

Core • 5-19 • 7 | 20 | 9,18 | 4 | 180 | 80 | yes | sometimes | 25 | 9 | 24 | 20
| 65 | 6,6 | 8

Challenge • 20-23 • 24 | 28 | 40 | 3